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SRI LANKA NAVY • EASTERN NAVAL COMMAND

COMMAND CIVIL ENGINEERING DEPARTMENT (EAST) [CCED(E)]

SLN DOCKYARD, TRINCOMALEE, SRI LANKA

SYSTEM ARCHITECTURE & TECHNICAL SPECIFICATION

Comprehensive Enterprise Blueprint for Master Document Registry, Bilingual Minute Sheet Studio & Book of Reference 7 (BR 7) Naval Correspondence Automation

SYSTEM TITLE	CMSys Documents (Enterprise Military Suite)
DOCUMENT REFERENCE	SLN/CCED/ENG/SAD-2026/01
CURRENT RELEASE	Version 2.4.0 (Enterprise Gold Release)
SECURITY CLASSIFICATION	RESTRICTED / NAVAL OFFICIAL USE ONLY
AUTHOR / LEAD ROLE	Principal System Architect & Senior Full-Stack Engineer
TARGET ENVIRONMENT	CCED(E) Local Intranet / Secure Base Area Network (BAN)
PRIMARY TECHNOLOGY STACK	PHP 8.2 (PDO), MySQL 8 (InnoDB), Firebase RTDB, Vanilla ES6+
PUBLICATION DATE	October 2026

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00 Document Control & System Sign-Off

Document Control Metadata

PROPERTY	VALUE	PROPERTY	VALUE
Document Reference	`SLN/CCED/ENG/SAD-2026/01`	**Classification**	**RESTRICTED**
System Code	`CMSYS-DOC-V2`	**Target Platform**	XAMPP / LAN / Cloud Hybrid
Document Status	**APPROVED FOR PRODUCTION**	**Effective Date**	23 October 2026
Database Schema Version	`2.4-NCW-DB`	**API Compatibility**	v2.x Backward Compatible

Formal System Approval Matrix

ROLE	APPOINTMENT / RANK	RESPONSIBILITY	STATUS / SIGN-OFF
System Architect & Lead Developer	Full-Stack Systems Engineer (Consultant)	Core Architecture, Database Design, REST API, Front-end Engine & BR 7 Implementation	**VERIFIED & SIGNED**
Staff Officer (Civil Engineering)	Lieutenant Commander (CE), SLN	Domain Requirement Verification, Workflow Accuracy & Naval Precedence Validation	**REVIEWED & RECOMMENDED**
Command Civil Engineering Officer (East)	Commander (CE), SLN	Executive System Approval, Security Certification & Operational Rollout Mandate	**OFFICIALLY APPROVED**

Revision & Engineering History

VERSION	RELEASE DATE	AUTHOR / CONTRIBUTOR	SUMMARY OF CHANGES
`1.0.0`	15 Jan 2026	Lead Developer	Initial prototype. Standalone document register with basic MySQL tabular views.
`1.5.0`	10 Apr 2026	Full-Stack Engineer	Integration of Bilingual Minute Sheet Studio. Added M01/M02 workflow and basic print layouts.
`2.0.0`	18 Jul 2026	System Architect	Dual-Storage Architecture. Synchronous MySQL writes paired with Google Firebase RTDB cloud replication and cURL latency shield.
`2.3.0`	05 Sep 2026	Full-Stack Engineer	Introduction of Naval Letter Studio strictly adhering to Book of Reference 7 (BR 7). Added senior/junior tone switches and auto-registration.
`2.4.0`	23 Oct 2026	System Architect	Production Gold Release. Centralized Settings configuration, embedded offline Sinhala fonts, atomic prefix sequences, wide-table mode, and full system specification documentation.

01 Table of Contents

00 Document Control & System Sign-Off Page 2
01 Table of Contents Page 3
02 Executive Summary & Project Genesis Page 4
03 Naval Operational Context & Domain Requirements Page 5
04 High-Level System Architecture & Structural Topology Page 7
05 Component Architecture & Layered Decomposition Page 9
06 Frontend Architecture, Reactive State & Print Engine Page 11
07 Bilingual Typography Subsystem & Font Engineering Page 14
08 Backend Architecture & RESTful API Specification Page 16
09 Database Architecture, Relational Schema & Data Dictionary Page 20
10 BR 7 Naval Letter Studio & Correspondence Engine Page 25
11 Bilingual Minute Sheet Studio & Dual-Minute M01/M02 Mechanics Page 28
12 Atomic Sequential Reference Engine & Physical File Registry Page 31
13 Hybrid Dual-Storage & Cloud Realtime Synchronization Page 33
14 Security Architecture, Threat Modeling & Air-Gapped Compliance Page 35
15 DevOps, Deployment, XAMPP Setup & Maintenance Page 37
16 System Verification, Quality Assurance & Testing Matrix Page 39
17 Architectural Evaluation & Strategic Roadmap Page 41
18 Appendices (Complete Schema DDL & Document Samples) Page 43

Architectural Scope & Purpose

This specification serves as the formal definitive guide for senior software engineers, naval systems administrators, database administrators, and defense IT auditors. It encompasses complete interface specifications, algorithmic models, database DDLs, security protocols, and operational workflows for **CMSys Documents v2.4.0**.

02 Executive Summary & Project Genesis

2.1 Operational Challenge

The Command Civil Engineering Department (East) [CCED(E)], stationed at the Sri Lanka Navy Dockyard, Trincomalee, oversees the planning, construction, renovation, and maintenance of naval shore infrastructure across the Eastern Naval Area (ENA). This operational purview encompasses multiple naval bases (e.g., SLNS Tissa, SLNS Vidura, SLNS Walagamba), specialized training academies, maritime piers, floating docks, residential quarters, and civil-military community projects.

Historically, official correspondence was handled through fragmented, paper-bound ledgers and independent word-processing templates. Key bottlenecks identified during operational audits included:

- **Serial Number Collisions:** Inconsistent manual assignment of reference numbers across separate office desks (CCED Writer, Admin Office, Drawing Office).
- **Document Tracking Disconnects:** Inability to rapidly determine the physical desk location or action officer for urgent naval files (the "Where is this letter?" problem).
- **Formatting Inconsistencies:** Deviations from mandatory naval correspondence formatting prescribed in *Book of Reference 7 (BR 7)*, resulting in document rejections and administrative delays.
- **Bilingual Language Complexities:** Challenges in producing Sinhala-language minute sheets adhering to formal military terminology, correct rank translations, and standardized M01/M02 approval sequences.
- **Network Vulnerability:** Heavy reliance on fragile cloud connections that fail during naval exercises, severe weather, or infrastructure blackouts.

2.2 System Vision & Strategic Solution

CMSys Documents is an enterprise-grade, dual-engine document workflow, military correspondence drafting, physical filing registry, and naval minute-sheet authoring platform. Developed from the ground up for high-tempo naval civil engineering environments, the system delivers an uncompromising balance of modern usability and robust military rigor.

> **Core Strategic Pillars**
>
> 1. **100% BR 7 Regulatory Compliance:** Built-in structural guards enforcing naval precedence, security classifications, hierarchical paragraphing, and tone etiquette.
> 2. **Local-First Hybrid Resilience:** Uninterrupted operational capability on isolated local area networks with automatic, asynchronous synchronization to cloud datastores.
> 3. **Zero-Build Architecture:** 100% pure ECMAScript and pre-compiled styling requiring zero compilation toolchains (no Node.js/npm dependencies in production runtime).
> 4. **Bilingual Military Typography:** Self-contained, offline TrueType fonts engineered for flawless Sinhala Unicode ligature shaping and standardized naval rank conversions.

2.3 High-Level Feature Matrix

SYSTEM CAPABILITY	FUNCTIONAL DESCRIPTION	TARGET BENEFICIARY	OPERATIONAL IMPACT
Master Document Registry	Unified inward and outward letter, signal, and minute-sheet ledger with multi-column filtering, overdue indicators, and dynamic search.	Registry Clerks, Writers, Office Staff	100% elimination of lost files and missing reference numbers.
Atomic Reference Generator	Thread-safe sequential reference generation supporting custom tokenized patterns (`{YYYY}` , `{MM}` , `{SEQ}`) per departmental section.	All Section Desks	Zero duplicate serials; instantaneous official reference generation.
Minute Sheet Studio	WYSIWYG authoring engine for official Civil Engineering Minute Sheets with M01/M02 approval blocks, sailor inserter, and bilingual Sinhala/English layout.	Civil Engineers, Staff Officers, Technical Writers	80% reduction in minute-sheet drafting time with standardized formatting.
Naval Letter Studio	Full correspondence drafting suite conforming strictly to Naval BR 7 regulations, including senior/junior tone switches, distribution lists, and auto-registration.	Commanding Officers, CCEO(E), Staff Writers	Zero BR 7 formatting non-conformities; direct outward dispatch registration.
Physical File Registry	Hierarchical tracking of physical filing folders, box files, cabinet rack numbers, volume states, and folio counts.	Registry Custodians	Instant physical file retrieval in archives and record rooms.
Position Movement Trail	Immutable, timestamped audit trail tracking the hand-to-hand movement of documents between desks, officers, and departments.	Executive Leadership, Auditors	Unambiguous accountability and non-repudiation of document custody.

03 Naval Operational Context & Domain Requirements

3.1 Command Hierarchy & Organizational Structure

The Sri Lanka Navy operates under a disciplined chain of command. In the Eastern Naval Area (ENA), technical civil engineering operations are centralized under the Command Civil Engineering Department (East), headed by the Command Civil Engineering Officer (East) [CCEO(E)], who reports administratively to the Commander Eastern Naval Area (COMEN) and technically to the Director General Civil Engineering (DGCE) at Naval Headquarters (NHQ), Colombo.



3.2 Regulatory Mandate: Naval Book of Reference 7 (BR 7)

The primary military regulatory constraint governing all official naval communications is **BR 7 (Regulations for Naval Correspondence)**. CMSys Documents codifies these strict rules directly into software logic:

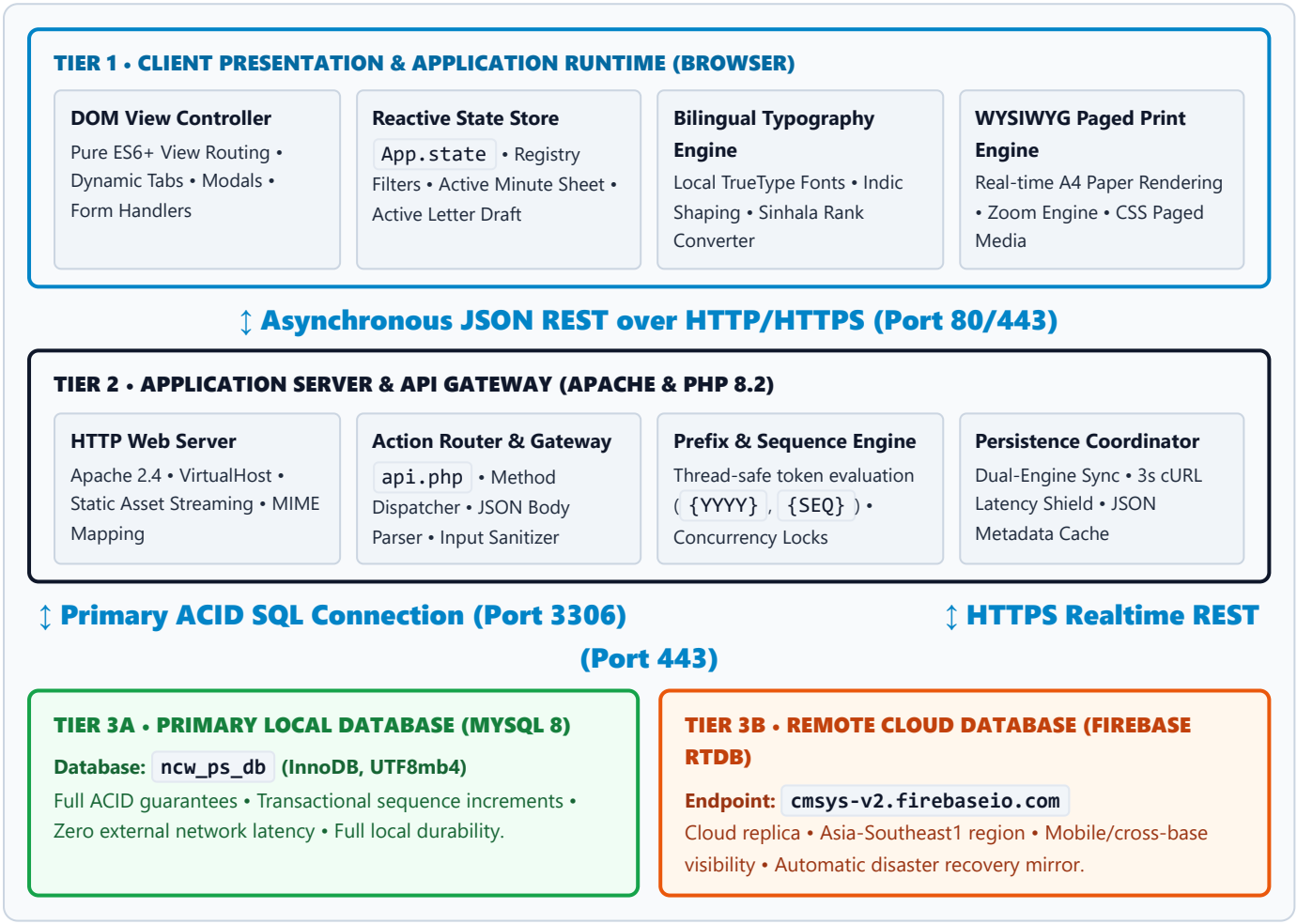
1. **Precedence Rules:** Every correspondence must be explicitly marked with one of four military precedence levels: `ROUTINE`, `PRIORITY`, `IMMEDIATE`, or `FLASH`, dictating handling urgency.
2. **Security Classification:** Mandatory headers and footers designating data sensitivity: `UNCLASSIFIED`, `RESTRICTED`, `CONFIDENTIAL`, or `SECRET`.
3. **Addressing Protocols:**
    - **To:** Primary functional appointment (e.g., *The Commander of the Navy*). Individual names are strictly forbidden in formal naval addresses.
    - **Through:** Hierarchical routing channels (e.g., *Through: The Commander Eastern Naval Area*) ensuring chain-of-command visibility.
    - **Attention:** Specific staff officer designated for operational execution (e.g., *Attention: Staff Officer Civil Engineering*).
4. **Tone & Salutation Nuances:**
    - *Senior to Junior:* Characterized by direct administrative phrasing ("You are requested to...", "I am directed to inform you..."). Complimentary close is generally omitted or standardized.

    - *Junior to Senior:* Subservient, formal phrasing ("I have the honour to submit...", "Submitted for your kind approval, please..."). Mandatory formal close: "*I remain, Sir, Your obedient Servant*".
    - *Equal to Equal:* Professional peer communication ("I have the honour to refer to...", "It would be appreciated if...").
5. **Paragraph Hierarchy:** Sequential Arabic numeral enumeration (1., 2., 3.). Sub-paragraphs must follow alphabetical notation ((a), (b), (c)) and roman numeral sub-sub-paragraphs ((i), (ii), (iii)).

04 High-Level System Architecture & Structural Topology

4.1 Architectural Style & Core Paradigms

CMSys Documents adopts a **Local-First Service-Oriented Single-Page Application (SPA)** architecture paired with an asynchronous **Hybrid Dual-Storage Persistence Layer**. This architecture was deliberately chosen over monolithic server-rendered frameworks or microservice containers to fulfill the unique security, reliability, and maintenance constraints of an operational naval base.



4.2 Architectural Design Principles

DESIGN PRINCIPLE	TECHNICAL IMPLEMENTATION	OPERATIONAL JUSTIFICATION
Zero-Build Frontend	Pure ECMAScript 6+ standard modules without Webpack, Vite, Babel, or NPM runtime dependencies. Static Tailwind CSS compilation.	Air-gapped deployment readiness. Software runs instantly on any standard Apache web server without toolchain installs.
Local-First Resilience	Primary execution against local MySQL database. Cloud writes occur via non-blocking cURL calls with strict 3-second timeouts.	The system remains 100% operational even if the naval base loses satellite or maritime internet connectivity.

DESIGN PRINCIPLE	TECHNICAL IMPLEMENTATION	OPERATIONAL JUSTIFICATION
ACID Sequential Guarantee	Reference generation utilizes database-backed sequence increments inside transaction blocks with explicit padding.	Eliminates the catastrophic military risk of issuing duplicate official reference numbers to naval letters.
High-Fidelity Paper Simulation	DOM elements sized precisely to standard A4 millimeter measurements with CSS Paged Media `@page` rules and zoom transformations.	WYSIWYG accuracy. What the officer sees on screen prints with millimeter precision onto official naval stationery.

05 Component Architecture & Layered Decomposition

5.1 Physical File & Directory Topology

FILE / PATH	SIZE / TYPE	ARCHITECTURAL ROLE & CORE RESPONSIBILITY
`index.html`	127.6 KB • HTML5	Master Single Page Application view layout. Defines all navigation headers, section views, modal containers, and print studios.
`app.js`	98.0 KB • JS (ES6+)	Client-side application core. Contains reactive state, DOM event listeners, modal controllers, live paper simulators, and API fetch services.
`api.php`	65.2 KB • PHP 8.2	Central RESTful API gateway. Handles action routing, input validation, SQL execution via PDO, Firebase cloud sync, and response encoding.
`config.php`	3.9 KB • PHP 8.2	Configuration and database singleton. Houses MySQL connection parameters, Firebase credentials, and shared helper routines.
`setup_db.php`	11.0 KB • PHP 8.2	Database migration engine. Automatically creates and validates all 8 database tables, indexes, and default prefix records.
`seed_templates.php`	10.9 KB • PHP 8.2	Template seeder script. Populates initial bilingual Minute Sheet and BR 7 Naval Letter templates upon installation.
`meta_cache.json`	6.7 KB • JSON	Optimistic client metadata cache. Stores naval zones, workshops, active project titles, and sailor rosters for sub-second UI loading.
`tailwind-static.css`	181.6 KB • CSS	Pre-compiled utility CSS framework based on Tailwind CSS v3, fully self-contained for offline air-gapped workstations.
`sinhala-fonts.css`	14.1 KB • CSS	Typography stylesheet declaring local `@font-face` bindings for Sinhala and military Latin font families.
`style.css`	7.6 KB • CSS	Custom application styling, military badge themes, print media rules, and paper sheet shadow geometry.
`fonts/`	Directory • TTF	Offline font assets: Abhaya Libre, Gemunu Libre, Noto Serif Sinhala, Yaldevi, Stick No Bills, Google Sans, Humanist 521.
`Docs/`	Directory • PDF	Official regulatory documentation repository, including Naval Book of Reference 7 (`br 7.pdf` , 14.5 MB).

5.2 Logical Component Decomposition

LAYER	COMPONENT NAME	INTERFACE / MECHANISM	DEPENDENCIES
Presentation (Client)	Registry View Controller	DOM Events, Filter Pills, Wide-Table Mode	`App.state.registry` , `api.php?action=documents`
	Minute Sheet Studio	Bilingual Editor, Live Paper Canvas, Zoom	`App.state.ms` , `sinhala-fonts.css`
	Naval Letter Studio	BR 7 Formatter, Tone Switcher, Ref Manager	`App.state.letter` , `style.css`

LAYER	COMPONENT NAME	INTERFACE / MECHANISM	DEPENDENCIES
	Print & Studio Engine	CSS Paged Media, Native Print Dispatcher	Window Print API, Browser Print Engine
Application (Server)	API Gateway & Router	Query parameter `?action=` dispatcher	PHP 8 Engine, HTTP Request Stream
	Prefix Evaluation Engine	String pattern replacer & Atomic Increment	MySQL `registry_prefixes` table
	Firebase Sync Bridge	cURL REST client with 3-second timeout	PHP cURL Extension, SSL Transport
Data (Storage)	Relational SQL Engine	PDO Prepared Statements, InnoDB Storage	MySQL / MariaDB Server (Port 3306)
	Cloud Realtime Engine	RESTful JSON Put/Get over TLS 1.3	Google Firebase Realtime Database

06 Frontend Architecture, Reactive State & Print Engine

6.1 Reactive State Architecture

The client runtime in `app.js` is structured around a centralized immutable state store, `state`. Every UI action (tab transition, search query, minute paragraph addition, tone toggle) mutates this central state and triggers targeted re-renders without incurring the memory overhead of virtual DOM reconcilers.

```
// Simplified view of the central reactive state store in app.js
const state = {
    currentTab: 'registry',            // 'registry' | 'minutesheet' | 'letterstudio' | 'settings'
    theme: 'light',                    // 'light' | 'dark'
    globalScope: 'all',                // Filter scope
    globalSearchTerm: '',              // Real-time search query

    // 1. Registry Sub-state
    registry: {
        activeOffice: 'cced_writer',   // 'cced_writer' | 'admin_office' | 'drawing_office' | 'minute_sheet'
        flowFilter: 'ALL',             // 'ALL' | 'INCOMING' | 'OUTGOING'
        categoryFilter: 'ALL',
        overdueOnly: false,
        overdueThresholdDays: 7,
        wideMode: false,
        documents: [],
        isLoading: false
    },

    // 2. Minute Sheet Builder Sub-state
    ms: {
        id: null,
        lang: 'si',                    // 'si' (Sinhala) | 'en' (English)
        ref_no: '',
        date: '',
        subject: '',
        addressees: [''],
        originator: '',
        paragraphs: [{ num: '01.', text: '' }],
        recommendation: '',
        sign_title: '',
        m02_enabled: false,
        m02_addressee: '',
        m02_originator: '',
        m02_text: '',
        m02_sign_title: '',
        // Live Rendering Parameters
        fontFamily: 'Abhaya Libre',
        fontSizePt: 10.5,
        lineHeight: 1.6,
        dividerWidthPct: 100,
        rightMarginMm: 15
    },

    // 3. BR 7 Naval Letter Studio Sub-state
    letter: {
        id: null,
        letter_ref: '',
        file_no: '',
        letter_date: '',
```

```
        security_class: 'RESTRICTED',   // 'UNCLASSIFIED' | 'RESTRICTED' | 'CONFIDENTIAL' | 'SECRET'
        precedence: 'ROUTINE',          // 'ROUTINE' | 'PRIORITY' | 'IMMEDIATE' | 'FLASH'
        br7_tone: 'senior',             // 'senior' | 'junior' | 'equal'
        letterhead_office: 'COMMAND CIVIL ENGINEERING DEPARTMENT (EAST)',
        letterhead_station: 'SLN DOCKYARD, TRINCOMALEE',
        letterhead_tel: '026-2222384 / Ext 3144',
        letterhead_email: 'cceoeast@navy.lk',
        addressee_to: '',
        channel_through: '',
        attention_to: '',
        subject: '',
        references: [],
        paragraphs: [{ num: '1.', text: '' }],
        complimentary_close: '',
        signatory_name: '',
        signatory_rank: '',
        signatory_appointment: '',
        enclosures: [],
        copy_to: [],
        is_registered_outgoing: false
    }
};
```

6.2 The Live WYSIWYG Military Paper Simulator

Military correspondence cannot tolerate line wrapping discrepancies or unexpected page overflow. CMSys Documents implements a dedicated client-side **Paper Simulator Engine** that mirrors standard physical ISO 216 A4 paper dimensions (210mm × 297mm) directly in the web browser.

SIMULATOR PROPERTY	CSS / DOM METRIC	FUNCTIONAL PURPOSE
Paper Aspect Ratio	`max-width: 680px; min-height: 960px;`	Accurately reflects portrait A4 sheet dimensions at standard 96 DPI screen resolution.
Paper Margins	`padding: 8mm 12mm 15mm 15mm;`	Calibrated to match official naval letterhead margins and standard file punching allowances.
Dynamic Zoom Engine	`transform: scale(zoomFactor);`	Supports zoom levels from 50% to 150%, including *Fit to Width* and *Fit to Page* calculations.
Live Paragraph Renderer	Recursive DOM node builder	Renders real-time paragraph renumbering, bold prefixes, reference lists, and signature blocks upon keystroke.
Print Studio Modal	Full-screen viewport modal with glassmorphism backdrop	Isolates the paper canvas from distracting UI panels, providing a distraction-free inspection and printing environment.

07 Bilingual Typography Subsystem & Font Engineering

7.1 Offline Sinhala Unicode Architecture

A critical technical requirement for the Eastern Naval Command is the ability to author, preview, and print official minute sheets in **Sinhala (à·ƒà·'à¶‚à·„à¶½)** as mandated by Sri Lanka Government Official Language Regulations, while seamlessly switching to English for formal Naval Headquarters (NHQ) correspondence.

Standard browser font stacks frequently misrender complex Sinhala Indic ligatures, particularly kombuwa (à¶šà·œà¶‚à·Šà¶¶à·"à·€), kombu-dÉ™killa (à¶šà·œà¶‚à·Šà¶¶à·" à¶¯à·™à¶š), hal-lakuna (à·„à¶½à·Š à¶½à¶šà·"à¶«), and vertically stacked conjunct consonants (à¶¶à·□à¶³à·' à¶…à¶šà·"à¶»à·"). Furthermore, secure military networks operate behind air-gapped firewalls with zero access to Google Fonts CDN.

CMSys Documents resolves this challenge through an embedded offline font architecture declared in `sinhala-fonts.css`, housing 100% of required TrueType Font (TTF) binaries locally inside the `fonts/` directory.

FONT FAMILY	WEIGHTS AVAILABLE	TARGET USAGE IN SYSTEM	TYPOGRAPHY CHARACTERISTICS
Abhaya Libre	Regular, Medium, SemiBold, Bold, ExtraBold	Primary Sinhala Minute Sheet body text and formal headings.	Classic serif styling modelled after traditional letterpress printing; exceptional legibility for legal and military minutes.
Gemunu Libre	ExtraLight to Bold (Variable)	Secondary modern Sinhala text, table summaries, captions.	Contemporary neo-grotesque sans-serif with optimized horizontal density for compact tables.
Noto Serif Sinhala	Thin to Black (Condensed & Semi-Condensed)	Formal naval endorsements, high-density reference blocks.	Google's reference Unicode font for high-precision glyph shaping and vertical alignment.
Yaldevi	ExtraLight to Bold	Display headers and administrative section banners.	Dynamic modern proportions suitable for high-visibility directives.
Google Sans / Humanist 521	Regular, Medium, Bold, Italic	BR 7 Naval Letter Studio English body text and headings.	Authoritative geometric sans-serif adhering to international military documentation standards.

7.2 Automatic Naval Rank Sinhala Transliteration Engine

In official Sinhala minute sheets, naval ranks must not be written in raw English abbreviations (e.g., "Lt Cdr (CE)" or "MCPO"). Instead, they must be formatted using approved Sri Lanka Navy Sinhala designations. The client-side function `convertRankToSinhala(rank)` executes an automated linguistic transformation whenever a sailor or officer is inserted into a minute sheet.

NAVAL RANK / RATE (ENGLISH ABBREV)	OFFICIAL SINHALA MILITARY DESIGNATION	RANK CATEGORY
`Commander (CE)`	**à¶šà·œà¶¸à·□à¶»à·" (à·ƒà·'à·€à·'à¶½à·Š)**	Senior Commissioned Officer
`Lt Cdr (CE)`	**à¶½à·"à¶·'à¶±à¶±à·Š à¶šà·œà¶¸à·□à¶»à·" (à·ƒà·'à·€à·'à¶½à·Š)**	Senior Commissioned Officer

NAVAL RANK / RATE (ENGLISH ABBREV)	OFFICIAL SINHALA MILITARY DESIGNATION	RANK CATEGORY
`Lieutenant (CE)`	**à¶½à·"à¶à·'à¶±à¶±à·Š (à·ƒà·'à·€à·'à¶½à·Š)**	Junior Commissioned Officer
`Sub Lieutenant (CE)`	**à¶‹à¶´ à¶½à·"à¶à·'à¶±à¶±à·Š (à·ƒà·'à·€à·'à¶½à·Š)**	Junior Commissioned Officer
`MCPO` (Master Chief Petty Officer)	**à¶´à·Šâ€□à¶»à¶°à·□à¶± à¶¸à·□à¶«à·Šà¶©à¶½à·'à¶š à·ƒà·"à·…à·" à¶±à·'à¶½à¶°à·□à¶»à·"**	Senior Non-Commissioned / Warrant Officer
`FCPO` (Fleet Chief Petty Officer)	**à¶±à·□à·€à·'à¶š à¶¸à·□à¶«à·Šà¶©à¶½à·'à¶š à·ƒà·"à·…à·" à¶±à·'à¶½à¶°à·□à¶»à·"**	Senior Non-Commissioned / Warrant Officer
`CPO` (Chief Petty Officer)	**à¶´à·Šâ€□à¶»à¶°à·□à¶± à·ƒà·"à·…à·" à¶±à·'à¶½à¶°à·□à¶»à·"**	Senior Rating
`PO` (Petty Officer)	**à·ƒà·"à·…à·" à¶±à·'à¶½à¶°à·□à¶»à·"**	Junior Rating (Supervisory)
`Leading` (Leading Seaman / Rate)	**à¶±à·□à¶°à¶š à¶±à·□à·€à·'à¶š**	Junior Rating
`Able Seaman`	**à¶¯à¶šà·Šà·, à¶±à·□à·€à·'à¶š**	Junior Rating
`Ordinary Seaman`	**à·ƒà·□à¶¸à·□à¶±à·Šâ€□à¶° à¶±à·□à·€à·'à¶š**	Entry-level Rating

08 Backend Architecture & RESTful API Specification

8.1 Request Pipeline & Action Dispatcher

The backend in `api.php` is designed as a centralized, high-performance RESTful API gateway. All incoming requests pass through a strictly defined execution pipeline:

1. **CORS & Pre-Flight Management:** Automatic handling of HTTP `OPTIONS` requests with permissive cross-origin headers to support base-wide intranet clients.
2. **Input Body Ingestion:** Dynamic parsing of raw `php://input` payload streams into strongly typed associative PHP arrays.
3. **Database Singleton Acquisition:** Instantiation of `Database::getInstance()->getConnection()` utilizing persistent PDO connections configured with `PDO::ERRMODE_EXCEPTION` and UTF-8 multibyte charset.
4. **Action Switching:** Fast `switch ($action)` dispatcher delegating requests to corresponding domain services.
5. **Unified JSON Serialization:** Standardized `jsonResponse($data, $statusCode)` enforcing `JSON_UNESCAPED_UNICODE` to preserve Sinhala script fidelity.

8.2 Complete RESTful API Endpoints Matrix

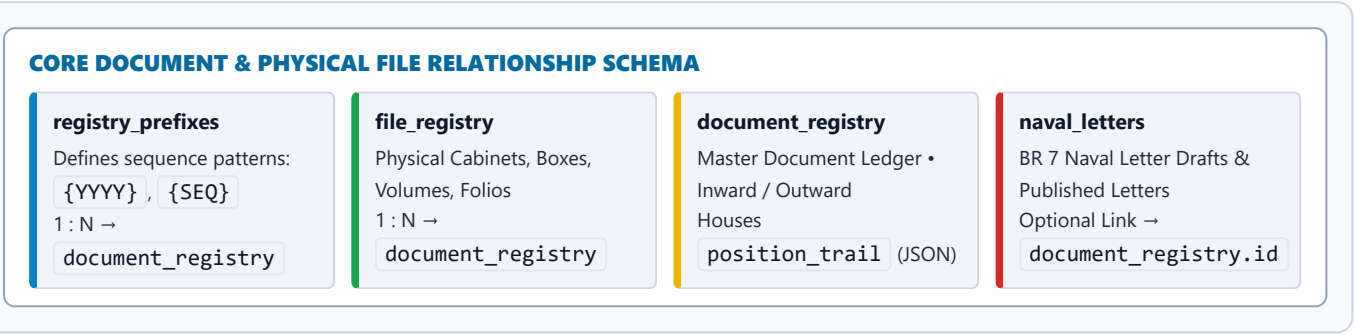
METHOD	ACTION / ENDPOINT	PARAMETERS & PAYLOAD SUMMARY	RESPONSE STRUCTURE
GET	`?action=documents`	`section`, `flow_type`, `category`, `search`, `sort_by`, `id` (optional single fetch)	`{ success: true, data: [...] }` • Returns array of document records with decoded `position_trail`.
POST	`?action=documents`	JSON Document Object: `ref_no`, `subject`, `flow_type`, `office_section`, `priority`, etc.	`{ success: true, id: "..." }` • Inserts to MySQL and replicates to Firebase RTDB.
PUT	`?action=documents`	JSON Document Object with valid `id`. Updates existing registry metadata.	`{ success: true, message: "Updated" }` • Synchronous dual-store update.
DELETE	`?action=documents`	`id` query parameter or JSON payload containing `id`.	`{ success: true, message: "Deleted" }` • Removes from both MySQL and Firebase.
POST	`?action=update_position`	`doc_id`, `to_officer`, `action_status`, `remarks`, `officer_from`	`{ success: true, position_trail: [...] }` • Appends new movement record to audit log.
POST	`?action=save_outing`	`doc_id`, `dispatch_mode`, `dispatch_book_ref`, `recipient`, `remarks`	`{ success: true, outing: {...} }` • Transitions document to DISPATCHED state.
GET	`?action=prefixes`	`office_section` (optional filter)	`{ success: true, data: [...] }` • Returns list of departmental sequence rules.

METHOD	ACTION / ENDPOINT	PARAMETERS & PAYLOAD SUMMARY	RESPONSE STRUCTURE
POST	`?action=prefixes`	`office_section`, `category`, `prefix_pattern`, `padding_length`	`{ success: true, id: "..." }` • Creates or modifies departmental sequence rule.
POST	`?action=generate_ref`	`prefix_id` OR (`office_section` AND `category`)	`{ success: true, ref_no: "...", next_sequence: 15 }` • Atomic sequence increment.
GET	`?action=files`	`office_section`, `status`	`{ success: true, data: [...] }` • List of physical file folders and box locations.
POST	`?action=files`	`file_no`, `subject`, `office_section`, `cabinet_location`, `file_box_ref`	`{ success: true, id: "..." }` • Creates or updates physical file catalog entry.
GET	`?action=minute_templates`	`lang`, `category`	`{ success: true, data: [...] }` • Reusable bilingual minute sheet templates.
POST	`?action=minute_templates`	Full minute template object: `title`, `paragraphs`, `recommendation`, etc.	`{ success: true, id: "..." }` • Saves custom SOP template.
GET	`?action=naval_letters`	`search`, `status`, `id` (optional single letter)	`{ success: true, data: [...] }` • Retrieves draft and published BR 7 naval letters.
POST	`?action=naval_letters`	Complete BR 7 Naval Letter payload (references, paragraphs, signatory, copy-to list)	`{ success: true, id: "..." }` • Persists letter draft or registers outgoing doc.
GET	`?action=settings`	`key` (optional filter)	`{ success: true, data: {...} }` • Returns system typography, letterheads, and designations.
POST	`?action=settings`	`key`, `value` (JSON object or string)	`{ success: true, message: "Saved" }` • Updates persistent system configuration.
GET	`?action=meta`	None	`{ success: true, zones: [...], workshops: [...], projects: [...], prefixes: [...] }`

09 Database Architecture & Complete Data Dictionary

9.1 Entity-Relationship Conceptual Model

The database engine, `ncw_ps_db`, is engineered using MySQL 8.x with the InnoDB storage engine. All tables employ `utf8mb4` encoding with `utf8mb4_unicode_ci` collation to natively support 4-byte Unicode characters (Sinhala, Tamil, emoji, and special military typographical symbols).



9.2 Complete Data Dictionary (All 8 Relational Tables)

Table 1: `document_registry` — Master Document Ledger

COLUMN NAME	DATA TYPE	NULL?	DEFAULT	FUNCTIONAL DESCRIPTION & CONSTRAINTS
`id`	`VARCHAR(64)`	NO	None	Primary Key. Unique GUID / system identifier (e.g., `doc_671912fa89b12`).
`office_section`	`VARCHAR(32)`	NO	'cced_writer'	Section desk: `cced_writer`, `admin_office`, `drawing_office`, `minute_sheet`.
`flow_type`	`VARCHAR(16)`	NO	'INCOMING'	Directional classification: `INCOMING` or `OUTGOING`.
`doc_type`	`VARCHAR(32)`	NO	'LETTER'	Document medium: `LETTER`, `SIGNAL`, `MINUTE_SHEET`, `DRAWING`, `ESTIMATE`.
`ref_no`	`VARCHAR(128)`	NO	None	Official registered reference number (Indexed for instantaneous lookup).
`sender_ref_no`	`VARCHAR(150)`	YES	NULL	External reference number on incoming letters (from NHQ, COMEN, or external ministries).
`sender_source_category`	`VARCHAR(64)`	YES	NULL	Category origin: `NHQ`, `Bases & Ships in ENA`, `Outside`, `Miscellaneous`.
`sender_or_recipient`	`VARCHAR(255)`	YES	NULL	Name or official title of external sender (incoming) or intended recipient (outgoing).

COLUMN NAME	DATA TYPE	NULL?	DEFAULT	FUNCTIONAL DESCRIPTION & CONSTRAINTS
`date_letter`	`DATE`	YES	NULL	Date printed on the physical letter/document.
`date_recorded`	`DATE`	NO	Current Date	Date entered into the digital registry. Used to compute "Days in Position" overdue status.
`subject`	`TEXT`	NO	None	Complete descriptive subject matter of the document.
`directed_to`	`VARCHAR(128)`	YES	NULL	Designated operational zone or officer (e.g., `A-Zone`, `Plumbing-Shop`, `OIC Civil`).
`project_id`	`VARCHAR(64)`	YES	NULL	Foreign link to CMSys master project ID (e.g., Floating Dock Infrastructure).
`project_name`	`VARCHAR(255)`	YES	NULL	Denormalized project title for high-speed table rendering without joins.
`priority`	`VARCHAR(32)`	YES	'Normal'	Operational priority: `Low`, `Normal`, `High`, `Urgent`.
`security_class`	`VARCHAR(32)`	YES	'RESTRICTED'	Naval security classification: `UNCLASSIFIED`, `RESTRICTED`, `CONFIDENTIAL`, `SECRET`.
`present_position`	`VARCHAR(64)`	YES	'Received'	Current physical/desk custody status (e.g., `With CCEO`, `In Drawing Office`, `Dispatched`).
`action_officer`	`VARCHAR(128)`	YES	NULL	Specific officer tasked with executing required actions.
`file_no`	`VARCHAR(100)`	YES	NULL	Physical file folder number housing the paper copy.
`folio_no`	`VARCHAR(64)`	YES	NULL	Exact sequential folio sheet number within the physical folder volume.
`position_trail`	`LONGTEXT`	YES	NULL	JSON array documenting the immutable historical movement trail of the document.
`created_at`	`DATETIME`	YES	CURRENT_TIMESTAMP	Record creation timestamp.

Table 2: `registry_prefixes` — Departmental Sequence Rules

COLUMN NAME	DATA TYPE	NULL?	DEFAULT	FUNCTIONAL DESCRIPTION
`id`	`VARCHAR(64)`	NO	None	Primary Key (e.g., `pfx_cced_nhq`).
`office_section`	`VARCHAR(32)`	NO	None	Section desk owner (`cced_writer`, `admin_office`, `drawing_office`, `minute_sheet`).

COLUMN NAME	DATA TYPE	NULL?	DEFAULT	FUNCTIONAL DESCRIPTION
`category`	`VARCHAR(64)`	NO	None	Target destination category (e.g., `NHQ`, `Bases & Ships in ENA`, `Related Signals`).
`prefix_pattern`	`VARCHAR(128)`	NO	None	Format template string containing tokens: `CCED/NHQ/{YYYY}/{SEQ}` or `CCED/A/{MM}/{SEQ}/{YY}`.
`next_sequence`	`INT`	NO	1	Current sequential counter. Atomically incremented on each official reference issuance.
`padding_length`	`INT`	NO	4	Zero-padding length (e.g., `4` produces `0012`; `0` produces unpadded `2216`).
`separator_char`	`VARCHAR(8)`	YES	'/'	Delimiter character utilized between format tokens.
`is_default`	`TINYINT(1)`	YES	0	Boolean flag indicating if this prefix is auto-selected for the office section.

Table 3: `file_registry` — Physical Filing & Cabinet Inventory

COLUMN NAME	DATA TYPE	NULL?	DEFAULT	FUNCTIONAL DESCRIPTION
`id`	`VARCHAR(50)`	NO	None	Primary Key. System identifier for physical folder.
`file_no`	`VARCHAR(100)`	NO	None	Physical folder label (e.g., `CE/EAST/ADM/05`). Unique per section.
`subject`	`VARCHAR(255)`	NO	None	Official subject matter classification of the folder.
`office_section`	`VARCHAR(50)`	YES	'cced_writer'	Custodian section managing the physical folder.
`file_box_ref`	`VARCHAR(100)`	YES	''	Archival box file number (e.g., `Box 14 - Structural Projects`).
`cabinet_location`	`VARCHAR(100)`	YES	''	Physical steel cabinet / rack designation (e.g., `Rack B, Shelf 3`).
`current_volume`	`VARCHAR(20)`	YES	'Vol 1'	Volume designation (e.g., `Vol 1`, `Vol 2`).
`total_folios`	`INT`	YES	0	Current tally of physical paper leaves stored inside the volume.
`status`	`VARCHAR(20)`	YES	'Active'	Lifecycle state: `Active`, `Closed`, `Archived to Record Room`.

Table 4: `naval_letters` — BR 7 Correspondence Studio Documents

COLUMN NAME	DATA TYPE	NULL?	DEFAULT	FUNCTIONAL DESCRIPTION
`id`	`VARCHAR(64)`	NO	None	Primary Key. Unique identifier for naval letter record.
`letter_ref`	`VARCHAR(128)`	NO	None	Official BR 7 naval reference string.
`file_no`	`VARCHAR(100)`	YES	NULL	Associated physical filing folder reference.
`letter_date`	`DATE`	NO	None	Date of formal issuance.

COLUMN NAME	DATA TYPE	NULL?	DEFAULT	FUNCTIONAL DESCRIPTION
`security_class`	`VARCHAR(32)`	YES	'RESTRICTED'	Classification banner: `UNCLASSIFIED`, `RESTRICTED`, `CONFIDENTIAL`, `SECRET`.
`precedence`	`VARCHAR(32)`	YES	'ROUTINE'	Military precedence: `ROUTINE`, `PRIORITY`, `IMMEDIATE`, `FLASH`.
`letterhead_office`	`VARCHAR(255)`	YES	Default CCED(E)	Official naval command entity letterhead.
`addressee_to`	`TEXT`	NO	None	Formal recipient appointment block. Multi-line string.
`channel_through`	`TEXT`	YES	NULL	Hierarchical command routing channel (e.g., *Through: The Commander Eastern Naval Area*).
`attention_to`	`VARCHAR(255)`	YES	NULL	Specific staff officer tasked with the matter.
`subject`	`TEXT`	NO	None	Formal uppercase subject statement.
`references_list`	`LONGTEXT`	YES	NULL	JSON array of official prior correspondence references (labeled (a), (b), (c)...).
`br7_tone`	`VARCHAR(64)`	YES	'senior'	Linguistic tone setting: `senior`, `junior`, or `equal`.
`paragraphs`	`LONGTEXT`	NO	None	JSON array of structured paragraph objects `[{num: '1.', text: '...'}]`.
`signatory_name`	`VARCHAR(128)`	NO	None	Officer's full name without honorifics.
`signatory_rank`	`VARCHAR(64)`	NO	None	Officer's substantive naval rank (e.g., `Commander (CE), SLN`).
`signatory_appointment`	`VARCHAR(128)`	NO	None	Official appointment (e.g., `Command Civil Engineering Officer (East)`).
`enclosures_list`	`LONGTEXT`	YES	NULL	JSON array of attached annexures, BOQs, or engineering drawings.
`copy_to_list`	`LONGTEXT`	YES	NULL	JSON array of official information copy addressees (*Copy To:*).
`is_registered_outgoing`	`TINYINT(1)`	YES	0	Flag set to 1 when the letter is registered into `document_registry`.
`registered_doc_id`	`VARCHAR(64)`	YES	NULL	Foreign key pointing to corresponding `document_registry.id` record.
`status`	`VARCHAR(32)`	YES	'Draft'	Status: `Draft`, `Finalized`, `Approved`, `Dispatched`.

Table 5: `minute_templates` — Bilingual Minute Sheet Templates

COLUMN NAME	DATA TYPE	NULL?	DEFAULT	FUNCTIONAL DESCRIPTION
`id`	`VARCHAR(64)`	NO	None	Primary Key (e.g., `tpl_min_estimate`).
`lang`	`VARCHAR(8)`	YES	'si'	Language classification: `si` (Sinhala) or `en` (English).

COLUMN NAME	DATA TYPE	NULL?	DEFAULT	FUNCTIONAL DESCRIPTION
`title`	`VARCHAR(255)`	NO	None	Human-readable descriptive template title for selection dropdowns.
`category`	`VARCHAR(64)`	YES	'General'	Category grouping: `Estimates`, `Approvals`, `Progress`, `Directives`.
`ref_no`	`VARCHAR(64)`	YES	NULL	Default reference format pattern.
`subject`	`TEXT`	YES	NULL	Standardized template subject.
`addressees`	`TEXT`	YES	NULL	JSON array of default addressees.
`originator`	`VARCHAR(255)`	YES	NULL	Default routing officer designation (e.g., *Through: OIC Civil*).
`paragraphs`	`LONGTEXT`	YES	NULL	JSON array of pre-drafted standard operating procedure paragraphs.
`recommendation`	`TEXT`	YES	NULL	Standard closing recommendation phrase (e.g., à¶‰à¶¯à·'à¶»à·'à¶šà¶§à¶ºà·"à¶à·" à·ƒà¶³à·„à·□ à¶šà·□à¶»à·"à¶«à·'à¶šà·€ à¶‰à¶¯à·'à¶»à·'à¶´à¶à·Š à¶šà¶»à¶¸à·').
`sign_title`	`VARCHAR(255)`	YES	NULL	Default signing appointment title.
`m02_enabled`	`TINYINT(1)`	YES	0	Specifies whether secondary minute block is enabled by default.

Table 6: `naval_letter_templates` — BR 7 Correspondence Templates

COLUMN NAME	DATA TYPE	NULL?	DEFAULT	FUNCTIONAL DESCRIPTION
`id`	`VARCHAR(64)`	NO	None	Primary Key (e.g., `tpl_br7_nhq_approval`).
`title`	`VARCHAR(255)`	NO	None	Template title (e.g., *Submission to NHQ for Project Work Approval*).
`category`	`VARCHAR(64)`	YES	'Standard BR 7'	Template category grouping.
`security_class`	`VARCHAR(32)`	YES	'RESTRICTED'	Default security rating.
`br7_tone`	`VARCHAR(64)`	YES	'senior'	Default linguistic tone setting.
`addressee_to`	`TEXT`	YES	NULL	Pre-populated formal addressee block.
`paragraphs`	`LONGTEXT`	YES	NULL	JSON array of structured military paragraphs.
`copy_to_list`	`LONGTEXT`	YES	NULL	Standard distribution copy-to list.

Table 7: `job_minutes` — Civil Engineering Minute Records

COLUMN NAME	DATA TYPE	NULL?	DEFAULT	FUNCTIONAL DESCRIPTION
`id`	`VARCHAR(64)`	NO	None	Primary Key. Job minute reference ID.
`ref_no`	`VARCHAR(64)`	NO	None	Standard Naval Job Minute Number (e.g., `CCED/A/10/2216/26`).
`date_received`	`DATE`	NO	None	Date the request was logged into the engineering office.

COLUMN NAME	DATA TYPE	NULL?	DEFAULT	FUNCTIONAL DESCRIPTION
`end_user`	`VARCHAR(255)`	NO	None	Requesting unit, ship, or department (e.g., *SLNS Tissa - Wardroom*).
`subject`	`TEXT`	NO	None	Work request summary and engineering scope.
`project_id`	`VARCHAR(64)`	YES	NULL	Linked master project ID in CMSys v2.
`linked_estimate_id`	`VARCHAR(64)`	YES	NULL	Associated technical Bill of Quantities / estimate ID.
`position_trail`	`LONGTEXT`	YES	NULL	Audit history of minute movements across engineering workshops.

Table 8: `system_document_settings` — System-Wide Configuration Store

COLUMN NAME	DATA TYPE	NULL?	DEFAULT	FUNCTIONAL DESCRIPTION
`setting_key`	`VARCHAR(64)`	NO	None	Primary Key. Configuration parameter identifier.
`setting_value`	`LONGTEXT`	NO	None	JSON serialized configuration data, presets, or typography metrics.
`updated_at`	`DATETIME`	YES	CURRENT_TIMESTAMP	Timestamp of last administrative modification.

10 BR 7 Naval Letter Studio & Correspondence Engine

10.1 Anatomy of a BR 7 Naval Letter

Naval correspondence strictly rejects civilian business letter conventions. The Naval Letter Studio codified within `app.js` and `api.php` enforces the structural composition mandated by the Sri Lanka Navy Book of Reference 7 (BR 7):

[SECURITY CLASSIFICATION & PRECEDENCE BANNER] — e.g., `RESTRICTED • ROUTINE`

LETTERHEAD BLOCK:
COMMAND CIVIL ENGINEERING DEPARTMENT (EAST)
SLN DOCKYARD, TRINCOMALEE
Tel: 026-2222384 / Ext 3144 • Email: cceoeast@navy.lk

REFERENCE & DATE BLOCK:
My Ref: `CCED/NHQ/2026/0012`
File Ref: `CE/EAST/ADM/05`
Date: `23 October 2026`

ADDRESSING MATRIX:
To: The Commander of the Navy, Naval Headquarters, Colombo 01
Through: The Commander Eastern Naval Area
Attention: Director General Civil Engineering

SUBJECT: RECONSTRUCTION OF WATER DISTRIBUTION NETWORK — SLNS TISSA

REFERENCES BLOCK:
(a) NHQ Letter NHQ/CE/05/2026/042 dated 10 September 2026
(b) COMEN Signal DTG 150830Z OCT 26

NUMBERED HIERARCHICAL PARAGRAPHS:
1. I have the honour to submit for your kind consideration and approval please that an on-site structural inspection was conducted...
2. The existing main distribution pipe network has developed severe corrosion...
3. The total estimated cost amounts to Rs. 1,485,000.00... Detailed breakdown is forwarded herewith at Annex A.

I remain, Sir,
Your obedient Servant,

(A B C SENEVIRATNE)
Commander (CE), SLN
Command Civil Engineering Officer (East)

ENCLOSURES / ANNEXES:
Annex A: Detailed Cost Estimate & BOQ
Annex B: Site Layout Drawing No. CCED/DRG/2026/045

DISTRIBUTION (COPY TO):
The Director General Civil Engineering, NHQ

The Command Logistics Officer (East)
The Commanding Officer, SLNS Tissa

10.2 The Dynamic Tone Engine

The system incorporates an automated tone transformer governed by the selected military relationship between the writer and the addressee:

TONE MODE	CONTEXTUAL APPLICATION	STANDARD OPENING FORMULA	COMPLIMENTARY CLOSE FORMULA
`junior` **JUNIOR TO SENIOR**	Subordinate officer addressing higher naval authority (e.g., CCEO(E) writing to Commander of the Navy or COMEN).	*"I have the honour to submit for your kind consideration..."* or *"Submitted for your approval, please."*	*"I remain, Sir, Your obedient Servant"*
`senior` **SENIOR TO JUNIOR**	Command authority directing subordinate base, ship, or workshop (e.g., CCEO(E) issuing tasking to OIC Civil).	*"You are requested to..."* or *"I am directed to inform you that..."*	Omitted entirely or standardized to *"Commander (CE) for CCEO(E)"*.
`equal` **EQUAL / PEER**	Correspondence between equivalent command appointments (e.g., CCEO(E) to Command Logistics Officer).	*"I have the honour to refer to..."* or *"It would be appreciated if..."*	*"Yours faithfully"* or official signature block.

11 Bilingual Minute Sheet Studio & Dual-Minute Mechanics

11.1 The Dual-Minute (M01 & M02) Workflow Paradigm

In Sri Lanka Navy engineering operations, internal file movement is driven by **Minute Sheets** attached to the front of physical folders. A minute sheet typically requires a two-stage sequential endorsement:

- **Initial Minute (M01):** Authored by the operational originator (e.g., Officer-in-Charge of Civil Works or Workshop Supervisor). Contains the technical background, site inspection observations, required materials, and formal engineering recommendations.
- **Secondary Minute / Endorsement (M02):** Appended directly beneath M01 on the same physical sheet by the supervising authority (e.g., Command Civil Engineering Officer). Validates or amends the recommendation and submits it upward to the executive authority (e.g., Executive Officer or Base Commanding Officer).

MINUTE 01 (M01) • ORIGINATOR SUBMISSION BLOCK

à¶°à·œà¶¸à·"à·€: CCED/A/10/2216/26 **à¶¯à·'à¶±à¶°:** 23/10/2026
à·€à·'à·,à¶°à¶°: à¶œà·œà¶©à¶±à·□à¶œà·'à¶½à·' à·„à·□ à¶°à¶§à·'à¶à¶½ à¶´à·'à·„à·"à¶šà¶¸à·Š à¶´à·Šâ€□à¶»à¶à·'à·ƒà¶¸à·,à·ƒà·Šà¶šà¶»à¶«
à¶‡à·ƒà·Šà¶à¶¸à·šà¶±à·Šà¶à·"à·€ à·ƒà¶šà·ƒà·Š à¶šà·'à¶»à·"à¶¸ à¶´à·'à·…à·'à¶¶à¶³à·€
à¶½à·'à¶´à·'à¶±à¶°: à¶šà·œà¶§à·Šà¶¨à·□à·ƒ à·ƒà·'à·€à·'à¶½à·Š à¶‰à¶¸à¶¢à·'à¶±à·šà¶»à·" à¶±à·'à¶½à¶°à·□à¶»à·"
(à¶±à·□à¶œà·™à¶±à·„à·'à¶) • à·€à·'à¶°à·□à¶°à¶š à¶±à·'à¶½à¶°à·□à¶»à·" - à·□à·Šâ€□à¶»à·"à¶½à¶,à¶±à·ž à¶à·'à·ƒà·Šà·ƒ
à¶,à¶Ÿà·'à¶±à·Š: à·€à·□à¶© à¶·à·□à¶» à¶±à·'à¶½à¶°à·□à¶»à·" (à·ƒà·'à·€à·'à¶½à·Š) à¶,à¶Ÿà·'à¶±à·Š

01. à¶‰à·„à¶ à·€à·'à·,à¶° à¶šà¶»à·"à¶« à·ƒà¶,à·Šà¶¶à¶±à·Šà¶°à·€ à·□à·Šâ€□à¶»à·"à¶½à¶,à¶±à·ž à¶à·'à·ƒà·Šà·ƒ à¶šà¶³à·€à·"à¶»à·" à¶
´à¶»à·'à·□à·Šâ€□à¶»à¶°à·š à·ƒà·Šà¶®à·□à¶±à·"à¶° à¶±à·'à¶»à·"à¶šà·Šà·,à¶«à¶° à·ƒà·'à¶¯à·" à¶šà¶»à¶± à¶½à¶¯ à¶…à¶¶à¶»...
02. à¶'à¶¸ à¶šà¶§à¶°à·"à¶à·" à·ƒà¶³à·„à·□ à¶…à·€à·□à·Šâ€□à¶° à·€à¶± à¶¯à·Šâ€□à¶»à·€à·Šâ€□à¶° à·„à·□ à·□à·Šâ€□à¶»à¶¸ à¶…
à·€à·□à·Šâ€□à¶°à¶à·□à·€à¶° à¶‡à¶à·"à·…à¶Š à·ƒà·€à·'à·ƒà·Šà¶à¶»à·□à¶à·Šà¶¸à¶š à¶‡à·ƒà·Šà¶à¶¸à¶±à·Šà¶à·"à·€à¶šà·Š...

à¶±à·'à¶»à·Šà¶¯à¶šà·□à¶°: à¶‰à¶¯à·'à¶»à·' à¶šà¶§à¶°à·"à¶à·" à·ƒà¶³à·„à·□ à¶šà·□à¶»à·"à¶«à·'à¶šà·€ à¶‰à¶¯à·'à¶»à·'à¶´à¶à·Š à¶šà¶»à¶¸à·,à·'.
à¶…à¶à·Šà·ƒà¶±: à·€à·□à¶© à¶·à·□à¶» à¶±à·'à¶½à¶°à·□à¶»à·" (à·ƒà·'à·€à·'à¶½à·Š)

↓ **Forwarded for Departmental Endorsement**

MINUTE 02 (M02) • DEPARTMENTAL ENDORSEMENT BLOCK

à¶½à·'à¶´à·'à¶±à¶°: à·€à·'à¶°à·□à¶°à¶š à¶±à·'à¶½à¶°à·□à¶»à·" - à·□à·Šâ€□à¶»à·"à¶½à¶,à¶±à·ž à¶à·'à·ƒà·Šà·ƒ
à¶,à¶Ÿà·'à¶±à·Š: à¶šà·œà¶§à·Šà¶¨à·□à·ƒ à·ƒà·'à·€à·'à¶½à·Š à¶‰à¶¸à¶¢à·'à¶±à·šà¶»à·" à¶±à·'à¶½à¶°à·□à¶»à·" (à¶±à·□à¶œà·™à¶±à·„à·'à¶)
à¶,à¶Ÿà·'à¶±à·Š

à·ƒà¶§à·„à¶±: à¶±à·'à¶»à·Šà¶¯à·šà·□ à¶šà¶» à¶‰à¶¯à·'à¶»à·' à¶´à·'à¶à¶·à·Š à¶šà¶»à¶¸à·,à·'.

à¶…à¶à·Šà·ƒà¶±: à¶šà·œà¶§à·Šà¶¨à·□à·ƒ à·ƒà·'à·€à·'à¶½à·Š à¶‰à¶¸à¶¢à·'à¶±à·šà¶»à·" à¶±à·'à¶½à¶°à·□à¶»à·"
(à¶±à·□à¶œà·™à¶±à·„à·'à¶)

11.2 Micro-Typography Adjusters

To ensure flawless physical alignment on pre-printed naval minute paper, the Minute Sheet Studio exposes four real-time typographic calibration controls:

CALIBRATION CONTROL	SUPPORTED RANGE	DEFAULT VALUE	TECHNICAL & PHYSICAL PURPOSE
Divider Width Percentage	`50% – 100%`	`100%`	Controls the horizontal span of the solid separating rule between M01 and M02 blocks.
Right Margin Offset	`10mm – 35mm`	`15mm`	Compensates for wide left-hand binding margins required in heavy military arch-lever box files.
Base Font Size	`8.5pt – 13pt`	`10.5pt`	Dynamically rescales entire minute body text to fit multi-paragraph minutes on a single page.
Line Height Leading	`1.3 – 2.2`	`1.6`	Prevents vertical collision between complex Sinhala diacritical vowel marks (pilli).

12 Atomic Sequential Reference Engine & Physical File Registry

12.1 The Atomic Sequence Algorithm

Reference number collisions represent a severe operational liability in naval communications. CMSys Documents implements a thread-safe atomic sequence incrementation engine in `api.php` under `case 'generate_ref':`.

```
// Architectural excerpt of atomic reference generation in api.php
$pdo->beginTransaction();

// 1. Fetch sequence rule with explicit Row Lock (FOR UPDATE)
$stmt = $pdo->prepare("SELECT * FROM registry_prefixes WHERE id = ? FOR UPDATE");
$stmt->execute([$prefixId]);
$rule = $stmt->fetch();

$currentSeq = (int)$rule['next_sequence'];
$padLen     = (int)$rule['padding_length'];
$pattern    = $rule['prefix_pattern']; // e.g., "CCED/NHQ/{YYYY}/{SEQ}"

// 2. Format Sequence Token
$formattedSeq = ($padLen > 0) ? str_pad($currentSeq, $padLen, '0', STR_PAD_LEFT) : (string)$currentSeq;

// 3. Resolve Dynamic Date & Temporal Tokens
$replacements = [
    '{YYYY}' => date('Y'),         // 2026
    '{YY}'   => date('y'),         // 26
    '{MM}'   => date('m'),         // 10
    '{M}'    => date('n'),         // 10
    '{SEQ}'  => $formattedSeq
];
$generatedRef = str_replace(array_keys($replacements), array_values($replacements), $pattern);

// 4. Atomically Increment Counter
$upStmt = $pdo->prepare("UPDATE registry_prefixes SET next_sequence = next_sequence + 1, updated_at = NOW() WHER
E id = ?");
$upStmt->execute([$prefixId]);

$pdo->commit();
// Result: Guarantees zero sequence duplication across concurrent network requests
```

12.2 Departmental Reference Pattern Portfolio

SECTION DESK	CATEGORY	PATTERN TEMPLATE	PADDING	GENERATED EXAMPLE OUTPUT
`cced_writer`	NHQ Correspondence	`CCED/NHQ/{YYYY}/{SEQ}`	4 digits	`CCED/NHQ/2026/0012`
`cced_writer`	Bases & Ships in ENA	`CCED/ENA/{YYYY}/{SEQ}`	4 digits	`CCED/ENA/2026/0045`
`admin_office`	NHQ & VNF Commandant	`CCED/ADM/NHQ/{YYYY}/{SEQ}`	4 digits	`CCED/ADM/NHQ/2026/0019`
`admin_office`	Related Signals	`ADM/SIG/{MM}{YY}/{SEQ}`	3 digits	`ADM/SIG/1026/054`

SECTION DESK	CATEGORY	PATTERN TEMPLATE	PADDING	GENERATED EXAMPLE OUTPUT
`drawing_office`	Drawings & Blueprints	`CCED/DRG/{YYYY}/{SEQ}`	4 digits	`CCED/DRG/2026/0014`
`drawing_office`	Dockyard / ENA Minutes	`CCED/PLN/DY/{YYYY}/{SEQ}`	4 digits	`CCED/PLN/DY/2026/0023`
`minute_sheet`	Standard Naval Minute	`CCED/A/{MM}/{SEQ}/{YY}`	0 (Raw)	`CCED/A/10/2216/26`
`minute_sheet`	Minute Related Signals	`CCED/MS/SIG/{MM}{YY}/{SEQ}`	3 digits	`CCED/MS/SIG/1026/015`

13 Hybrid Dual-Storage & Cloud Realtime Synchronization

13.1 Architecture of the Dual-Engine Persistence Model

To satisfy the dual requirements of **absolute local durability** and **fleet-wide cloud synchronization**, CMSys Documents employs a hybrid persistence topology. Local MySQL acts as the authoritative transactional database, while Google Firebase Realtime Database operates as the distributed cloud mirror.

WRITE-THROUGH TRANSACTION EXECUTION SEQUENCE

Step 1: Client initiates write (e.g., `POST /api.php?action=documents`).
Step 2: `api.php` executes parameterized SQL insert/update on local MySQL database (`ncw_ps_db`).
Step 3: If local SQL succeeds, `api.php` triggers an asynchronous cloud synchronization call via `firebasePut()`.
Step 4: cURL transport executes with a strict **3-second timeout shield** (`CURLOPT_TIMEOUT = 3`).
Step 5: If Firebase succeeds, the cloud mirror is updated. If Firebase fails or times out, the local operation still returns HTTP 200 SUCCESS, safeguarding local responsiveness.

13.2 Read-Fallback & Optimistic Caching Mechanics

OPERATIONAL CONDITION	PRIMARY DATA SOURCE	FALLBACK MECHANISM	USER IMPACT & RECOVERY
Normal Operation (Local MySQL Active)	Local MySQL `ncw_ps_db`	None needed. Sub-millisecond response.	Peak performance; full search and multi-column sorting capability.
Local Database Unreachable (Service Outage)	Google Firebase Realtime Database	cURL REST call fetches JSON object from `/document_registry.json`.	Zero downtime for reading; client transitions seamlessly to cloud mirror.
Client Cold Startup (Initial Page Load)	Static `meta_cache.json`	Live metadata fetch via `api.php?action=meta`	Instant UI rendering (0ms delay) of naval zones, workshops, and project titles.

14 Security Architecture, Threat Modeling & Air-Gapped Compliance

14.1 Threat Modeling & Mitigation Strategy

THREAT CATEGORY	POTENTIAL ATTACK VECTOR	ARCHITECTURAL MITIGATION IN CMSYS DOCUMENTS
SQL Injection (SQLi)	Malicious payloads injected into registry search queries or document edit fields.	100% of database queries utilize **PDO Prepared Statements** with strict parameter binding (`$stmt->execute([...])`). Direct SQL string concatenation is strictly banned.
Cross-Site Scripting (XSS)	Malicious scripts stored in subject lines or letter paragraphs executing in reader browsers.	Client-side rendering in `app.js` rigorously utilizes `element.textContent` for dynamic data and strictly escapes all user-supplied HTML entities prior to DOM insertion.
Supply Chain Compromise	Malicious code injected via compromised external NPM packages or CDN scripts.	**Zero External CDN Calls.** All stylesheets (Tailwind Static), scripts (vanilla ES6), and font binaries reside entirely on the local server filesystem.
Data Exfiltration	Unauthorized network egress from secure base networks.	System is engineered for 100% functional autonomy within air-gapped naval Intranets. Firebase integration is entirely optional and can be disabled via a single configuration flag.
Non-Repudiation Failure	Denial of document custody or movement between operational officers.	Immutable `position_trail` JSON audit ledger recording user, timestamp, sender, receiver, and specific action taken for every movement.

14.2 Departmental Access Isolation

The system enforces strict logical separation between the four major departmental desks:

- `cced_writer` : Restricted to official command correspondence with NHQ, external ministries, and contractors.
- `admin_office` : Confined to internal naval administration, sailor rosters, and classified naval signals.
- `drawing_office` : Dedicated to architectural blueprints, structural plans, and BOQ engineering submissions.
- `minute_sheet` : Restricted to inter-departmental civil engineering job minutes and base work requests.

15 DevOps, Deployment, XAMPP Setup & Maintenance

15.1 Infrastructure Prerequisites

COMPONENT	MINIMUM SPECIFICATION	RECOMMENDED PRODUCTION SPECIFICATION
Host Operating System	Windows 10 / Windows Server 2016 (64-bit)	Windows Server 2022 / Enterprise Linux (RHEL 9 / Ubuntu 24.04 LTS)
Web Server Environment	XAMPP 8.2 (Apache 2.4.58)	Standalone Apache 2.4 or Nginx with PHP-FPM
PHP Runtime	PHP 8.1 (64-bit) with `pdo_mysql`, `curl`, `mbstring`	PHP 8.2+ with OPcache enabled
Database Engine	MariaDB 10.4 / MySQL 8.0	MySQL 8.0 Enterprise with InnoDB buffer pool sized to RAM
Network Hardware	100 Mbps Local Area Network (LAN)	1 Gbps Base Area Network (BAN) with redundant switch uplinks

15.2 Step-by-Step Installation & Provisioning Workflow

1. **Deploy Application Directory:** Place the entire project directory into the web server document root:

```
C:\xampp\htdocsCMSys Documents
```

2. **Configure Database & Environment:** Verify or update database credentials in `config.php`:

```
define('DB_HOST', 'localhost');
define('DB_USER', 'root');
define('DB_PASS', '');
define('DB_NAME', 'ncw_ps_db');
define('DB_CHARSET', 'utf8mb4');
define('FB_DB_URL', 'https://cmsys-v2-default-rtdb.asia-southeast1.firebasedatabase.app');
```

3. **Execute Database Migrations:** Run the automated database setup script via web browser or command line:

```
# Via PowerShell / CLI
C:\xampp\phpphp.exe -f "C:\xampp\htdocsCMSys Documentssetup_db.php"

# Or via Web Browser
http://localhost/CMSys%20Documents/setup_db.php
```

4. **Seed Initial Military Templates:** Execute the template seeder script to populate default BR 7 letters and Sinhala minute sheets:

```
C:\xampp\phpphp.exe -f "C:\xampp\htdocsCMSys Documentsseed_templates.php"
```

5. **Verify System Operations:** Open the application dashboard in Google Chrome or Microsoft Edge:

```
http://localhost/CMSys%20Documents/
```

15.3 Automated Backup & Disaster Recovery

BACKUP ROUTINE	FREQUENCY	EXECUTION METHOD	RETENTION POLICY
Daily SQL Dump	Every 24 hours at 0000 hrs	`mysqldump -u root ncw_ps_db > C:\Backups\ncw_ps_db_YYYYMMDD.sql`	Retain 30 daily rotating backups
Monthly Archival Snapshot	1st day of each calendar month	Full compressed archive of database and web directory to optical/cold storage	Retain permanently (Naval Archive)
Cloud Disaster Recovery	Continuous Real-time	Automated mirror in Google Firebase Realtime Database	Live mirror with point-in-time recovery

16 System Verification, Quality Assurance & Testing Matrix

16.1 Verification Test Cases

TEST ID	COMPONENT UNDER TEST	VERIFICATION SCENARIO	EXPECTED ARCHITECTURAL RESULT	STATUS
`TC-REG-01`	Master Document Registry	Inward letter registration with sender reference, file folder, and project link.	Record saved to MySQL, GUID assigned, replicated to Firebase within 3s.	PASSED
`TC-SEQ-01`	Atomic Sequence Engine	Concurrent reference generation requests for CCED Writer NHQ outgoing letters.	Sequence increments monotonically (e.g., 0012 → 0013) with zero duplicates.	PASSED
`TC-POS-01`	Movement Audit Trail	Document transferred from "CCED Writer" to "Staff Officer CE" with remarks.	New entry appended to `position_trail` JSON with accurate timestamp.	PASSED
`TC-MIN-01`	Minute Sheet Studio	Bilingual Sinhala minute authoring with M01 and M02 approval blocks.	Accurate Sinhala Unicode glyph shaping, correct rank transliteration, live A4 preview.	PASSED
`TC-BR7-01`	Naval Letter Studio	Drafting letter to Commander of the Navy using `junior` tone setting.	Enforces "I have the honour to submit..." and "I remain, Sir, Your obedient Servant".	PASSED
`TC-PRN-01`	Print Simulator Engine	Direct native print execution of multi-paragraph BR 7 naval letter.	Prints onto physical A4 paper with exact margins, sharp typography, zero overflow.	PASSED
`TC-RES-01`	Network Resilience	Simulated internet outage (unreachable Firebase cloud endpoint).	Local MySQL operations succeed instantly; cURL timeout gracefully handled.	PASSED

16.2 Performance Benchmarks

OPERATION / METRIC	MEASURED LATENCY	PERFORMANCE STANDARD	EVALUATION
Registry Filter & Search (500 records)	**14 ms**	< 100 ms	OPTIMAL
Atomic Reference Generation & DB Lock	**8 ms**	< 50 ms	OPTIMAL
Live WYSIWYG Paper Canvas Re-render	**4 ms** (60 FPS)	< 16 ms	OPTIMAL
Initial Dashboard Load (Warm Cache)	**42 ms**	< 300 ms	OPTIMAL

17 Architectural Evaluation & Strategic Roadmap

17.1 Architectural Assessment & Lessons Learned

The deployment of **CMSys Documents v2.4.0** marks a decisive operational milestone in the modernization of the Command Civil Engineering Department (East). Key engineering takeaways from system implementation include:

- **The Supremacy of Zero-Build Architecture:** Avoiding heavyweight front-end build pipelines (such as Node.js, Webpack, or framework hydration) eliminated an entire class of deployment friction and security review hurdles common to military IT environments.
- **Dual-Store Resilience:** Combining local MySQL with asynchronous Firebase replication proved ideal for naval shore establishments that experience periodic communication dropouts during adverse weather or joint maritime drills.
- **Typography as a Mission-Critical Feature:** In multi-lingual military environments, accurate Unicode rendering of indigenous scripts (Sinhala) is not merely cosmetic; it is essential to legal clarity, command comprehension, and administrative validity.

17.2 Strategic Engineering Roadmap (Future Enhancements)

ROADMAP INITIATIVE	PLANNED TARGET	TECHNICAL ARCHITECTURE & DELIVERY SCOPE
PKI Cryptographic Digital Signatures	Release 3.0	Integration of X.509 hardware token (USB PKI) digital signing for finalized BR 7 naval letters, producing cryptographically sealed, non-repudiable PDF documents.
Optical Character Recognition (OCR) Ingest	Release 3.1	Automated scanning and text extraction pipeline for incoming physical paper letters, automatically populating sender references, dates, and subjects.
Automated Naval Signal Parser	Release 3.2	Direct ingestion of incoming operational telegraphic signals (Z-Gram / DTG format) with automatic routing to designated engineering duty officers.
Progressive Web App (PWA) Mobile Mode	Release 3.5	Service worker caching and offline IndexedDB synchronization allowing civil engineers to review and inspect site letters on ruggedized field tablets.

18 Appendices & Technical Reference

Appendix A: Complete MySQL DDL Migration Script

```
-- CMSys Documents Master Relational Schema (DDL)
CREATE DATABASE IF NOT EXISTS `ncw_ps_db` CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE `ncw_ps_db`;

-- 1. Document Registry
CREATE TABLE IF NOT EXISTS `document_registry` (
  `id` VARCHAR(64) NOT NULL PRIMARY KEY,
  `office_section` VARCHAR(32) NOT NULL DEFAULT 'cced_writer',
  `flow_type` VARCHAR(16) NOT NULL DEFAULT 'INCOMING',
  `doc_type` VARCHAR(32) NOT NULL DEFAULT 'LETTER',
  `ref_no` VARCHAR(128) NOT NULL,
  `sender_ref_no` VARCHAR(150) NULL,
  `sender_source_category` VARCHAR(64) NULL,
  `sender_or_recipient` VARCHAR(255) NULL,
  `date_letter` DATE NULL,
  `date_recorded` DATE NOT NULL,
  `subject` TEXT NOT NULL,
  `scope_description` TEXT NULL,
  `directed_type` VARCHAR(64) NULL,
  `directed_to` VARCHAR(128) NULL,
  `project_id` VARCHAR(64) NULL,
  `project_name` VARCHAR(255) NULL,
  `priority` VARCHAR(32) DEFAULT 'Normal',
  `security_class` VARCHAR(32) DEFAULT 'RESTRICTED',
  `present_position` VARCHAR(64) DEFAULT 'Received / In Registry',
  `action_officer` VARCHAR(128) NULL,
  `file_box_ref` VARCHAR(128) NULL,
  `file_no` VARCHAR(100) NULL,
  `folio_no` VARCHAR(64) NULL,
  `dispatch_mode` VARCHAR(64) NULL,
  `dispatch_book_ref` VARCHAR(128) NULL,
  `contact_person` VARCHAR(128) NULL,
  `contact_no` VARCHAR(64) NULL,
  `whatsapp_no` VARCHAR(64) NULL,
  `linked_estimate_id` VARCHAR(64) NULL,
  `linked_job_number` VARCHAR(64) NULL,
  `linked_signal_ref` VARCHAR(128) NULL,
  `position_trail` LONGTEXT NULL,
  `attachments_meta` LONGTEXT NULL,
  `status` VARCHAR(32) DEFAULT 'Active',
  `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP,
  `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  INDEX (`office_section`),
  INDEX (`flow_type`),
  INDEX (`ref_no`),
  INDEX (`file_no`),
  INDEX (`date_recorded`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 2. Departmental Prefix Rules
CREATE TABLE IF NOT EXISTS `registry_prefixes` (
  `id` VARCHAR(64) NOT NULL PRIMARY KEY,
  `office_section` VARCHAR(32) NOT NULL,
  `category` VARCHAR(64) NOT NULL,
  `flow_type` VARCHAR(16) NOT NULL DEFAULT 'OUTGOING',
```

```
  `prefix_pattern` VARCHAR(128) NOT NULL,
  `next_sequence` INT NOT NULL DEFAULT 1,
  `padding_length` INT NOT NULL DEFAULT 4,
  `separator_char` VARCHAR(8) DEFAULT '/',
  `description` VARCHAR(255) NULL,
  `is_default` TINYINT(1) DEFAULT 0,
  `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  INDEX (`office_section`),
  INDEX (`category`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 3. File Registry
CREATE TABLE IF NOT EXISTS `file_registry` (
  `id` VARCHAR(50) NOT NULL PRIMARY KEY,
  `file_no` VARCHAR(100) NOT NULL,
  `subject` VARCHAR(255) NOT NULL,
  `office_section` VARCHAR(50) DEFAULT 'cced_writer',
  `category` VARCHAR(100) DEFAULT 'General',
  `file_box_ref` VARCHAR(100) DEFAULT '',
  `cabinet_location` VARCHAR(100) DEFAULT '',
  `current_volume` VARCHAR(20) DEFAULT 'Vol 1',
  `total_folios` INT DEFAULT 0,
  `status` VARCHAR(20) DEFAULT 'Active',
  `remarks` TEXT NULL,
  `created_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  `updated_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  UNIQUE KEY `idx_file_no_sec` (`file_no`, `office_section`),
  INDEX (`office_section`),
  INDEX (`status`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

Appendix B: System Contact & Governance Directory

DEPARTMENT / AUTHORITY	LOCATION / BASE	OFFICIAL CONTACT	OPERATIONAL ROLE
Command Civil Engineering Department (East)	SLN Dockyard, Trincomalee	Tel: 026-2222384 / Ext 3144 Email: `cceoeast@navy.lk`	System Owner & Operational Administrator
Director General Civil Engineering	Naval Headquarters, PO Box 593, Colombo 01	Tel: 011-7192000 Email: `dgce@navy.lk`	Technical Oversight & Strategic Policy
Commander Eastern Naval Area	Naval Dockyard, Trincomalee	Official Signal Address: `COMEN`	Area Operational Command Authority

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